

Statistical Arbitrage in Cryptocurrencies

Momentum, reversal and funding carry on Binance data, 2020 to 2026

Summary

I tested momentum, reversal and funding-carry strategies on Binance data from 2020 to September 2026, charged realistic costs (20 bps for market orders, 7 bps for limit orders) and combined the ones that passed into a book. Parameters were chosen on a development window (2020 to July 2024) and screened on a gate window (August 2024 to June 2025), which also decided between configs that passed both, and the final book was run once on a lockbox year (July 2025 to July 2026). Every configuration the research chose between is in a registry, 53 rows in all: 48 research configurations, which the deflated Sharpe charges for, the book's two combination rules and v2's three rows. The post-hoc checks of section 5 and appendix A re-score frozen strategies and log nothing. The two baselines kept parameters set before the registry existed (section 2), and untilted momentum is logged twice, as the baseline and as one of Seasonality's configurations, so the 48 research rows hold 47 distinct strategies.

On the research data the book made money on its lockbox: a walk-forward Sharpe of 1.46 with a beta to BTC of -0.02. I then rebuilt the data from Binance's public archive to test that result, and found that the research coin list was mostly survivors. It was the 150 USDT pairs with the most 24-hour volume on 2026-07-06, and 46% of the universe's coin-days before then were coins it left out. With every pair included, the frozen book's lockbox Sharpe falls to 0.35 (walk-forward) and -0.62 (equal weight), so most of the result came from the coin list. That is the main lesson of the project: a coin list picked by recent volume is a list of survivors, and splitting the history in time does not catch it, because every window, the lockbox included, is drawn from the same list. Fills and market impact tell the same story for Orderflow, the sleeve that looked best.

The clearest positive result is funding carry: weighted by rank and measured on perp prices, it holds up on the full universe, with a Sharpe of 1.68 on dev and 2.42 on the gate at about 15% volatility. Dev and gate had both been seen when it was designed, so this is not out-of-sample evidence. A second version of the book built around it (v2) was registered in the repository before it was tested, and it is a hypothesis under test, not a validated strategy: its test is every day from 2026-10-02, recorded month by month in notebook 12.

The main results at a glance

Sharpe ratio	Dev	Gate	Lockbox	Forward
Frozen book, walk-forward, research coin list	+0.40	+2.52	+1.46	-1.01
Frozen book, walk-forward, every pair	-0.20	+1.85	+0.35	-0.90
Frozen book, equal weight, research coin list	+1.68	+2.90	+1.23	+0.63
Frozen book, equal weight, every pair	+0.79	-1.44	-0.62	+1.26
v2 book, every pair	+1.35	+1.13	not run	from 2026-10-02
v2 Carry sleeve, every pair	+1.68	+2.42	not run	from 2026-10-02

Every pair: the daily top 100 chosen from all Binance pairs, delisted ones included (section 5). The forward window is 2026-07-07 to 2026-09-26 (section 6). The walk-forward book needs 756 days of history, so its dev figure covers 2021-10 to 2024-07; over those dates the equal-weight book's is +0.46. v2 is never evaluated on the lockbox or on that forward window, since both had been seen before it was written. Its figures include two fixes made after its first run and before any test result was computed; as first run they were +1.39 and +1.12 for the book and +1.76 and +2.41 for Carry (section 7).

What the project shows

- Survivorship, the main lesson. The daily data was rebuilt from Binance's public archive for 585 coins, delisted ones included, and its prices and funding rates match the research data exactly where the two overlap. It showed that 46% of the historical universe was missing from the research coin list, and that the book's lockbox result came mostly from that gap (section 5).
- Funding carry, the clearest positive result: weighted by rank on perp prices, it holds up on the full universe on

dev and gate, with a Sharpe of 1.68 and 2.42. Both windows had been seen when it was designed, so v2, the book built on it, is a hypothesis under test on new data, not a validated strategy (section 7).

- A protocol that can fail. The history is split into development, gate and lockbox windows in time order, every configuration the research chose between is logged (53 rows, the 48 research ones charged for with deflated Sharpe ratios), the lockbox has one recorded run, and v2's rules were committed before any v2 number existed.

- Execution measured, not assumed. A fill model shows that 99% of limit orders fill, but the ones that miss are the days the price runs away, which costs Orderflow 43% of its price P&L on dev. A per-coin market-impact model shows its edge is mostly gone by \$1M (section 5).

- Code that checks itself. The shared library has tests for look-ahead, dollar neutrality, costs, the fill model and the data pipeline; the report is rebuilt from the notebooks' saved results and stops before writing the PDF if a headline number, in the report or the README, disagrees with them; and check.py runs every check in one command.

1. Data and method

- Daily klines for the top 150 Binance USDT pairs by volume, 2018-01 to 2026-07 (149 names returned data, 11 of which had stopped trading before the fetch). Close, USD volume and taker-buy volume are kept.

- Hourly klines for the 60 names with the highest 30-day volume at fetch time, 2020-01 to 2026-07. The hourly research starts in 2020-06.

- Perpetual funding rates for the same 60 names where a perp trades under the coin's own name (55 names; the others are four pegged assets and PEPE, whose perp is 1000PEPE), summed per day, from 2019-09.

- The data was fetched on 2026-07-06, so that last day is a partial day in every panel.

- For section 5 and v2, the same daily data for every Binance USDT pair that ever traded, from the public archive (585 coins once stablecoins, wrapped coins, pegged assets, tokenized stocks and leveraged tokens are removed), hourly bars for the coins that were ever in the top 100, and prices and funding for every perp.

The tradable universe is point-in-time: on each day, the 100 coins with the highest 30-day median dollar volume, using data up to the day before, with a \$1M floor and at least 30 days of history. Stablecoins, wrapped coins and leveraged tokens are excluded. The median universe since 2020 has 66 names. The 2018-2019 data serves as history for the first signals, and from September 2019 the sleeves' returns also train the first walk-forward weights.

The backtest applies weights set at the close of day t to the return of day $t+1$, so no position uses information it could not have had, with two exceptions. The pairs baseline picked its pairs with the close of the day whose return they then earned (section 5.6). And the coin lists themselves, the daily one and the 60 names with hourly bars and funding, were picked by volume on the fetch date; section 5 shows that most of the research result came from that choice. Costs are charged on turnover every day: 20 bps per dollar traded for market orders (7 bps commission plus 13 bps slippage) and 7 bps for limit orders. The momentum sleeves take liquidity and pay 20 bps; orderflow, carry and pairs rebalance passively and pay 7 bps. Backtests are unconstrained, and Sharpe ratios are annualised with 365 days, since crypto trades every day; for the 252-day convention used for stocks, multiply by 0.83 (the square root of 252/365), so the book's lockbox Sharpe of 1.46 becomes 1.21.

Validation windows

Window	Dates	Used for
Development	2020-01-01 to 2024-07-31	choosing signals and parameters
Gate	2024-08-01 to 2025-06-30	a check before anything is kept, and the pick between configs that pass
Lockbox	2025-07-01 to 2026-07-06	one final run of the frozen book

The windows split the history roughly 70/15/15 in time, the usual train, validation and test split. The gate also decided between configs that passed both windows, so the chosen sleeves' gate numbers are not out of sample; the lockbox is. The lockbox has one recorded run, in notebook 06, made after every selection decision; the project's first version had already run the two baselines over it (section 2). Appendix A and section 5 re-score the same frozen book with corrected inputs and say so.

2. Baselines: momentum and pairs

Two strategies serve as baselines that every later idea has to beat. Time-series momentum holds each coin long or short by the sign of its 30-day return (skipping the latest day) and scales the book to 15% volatility. The pairs sleeve re-selects up to 20 correlated pairs every 63 days, opens a spread when its z-score passes 2 and holds it until the z-score passes 2 on the other side. Their settings were fixed before the gate window, so the gate shows how they hold up on data they were not chosen on.

Those settings come from the project's first version, built on CoinGecko data from 2020-06-29 to 07-02, before this protocol existed. It picked them from small grids on 2020-2022 data and also tested holding the market only at weekends: 26 configurations by its own count, and about ten exploratory variants besides, none of them in the registry. It also ran both baselines as twsq alphas up to 2026-07-02, momentum from 2024-08 and pairs from 2025-07 (Sharpe -0.29 and -1.11), so for these two the lockbox was not unseen, and the plan for this protocol, written before its data was fetched, started from knowing that momentum had gone flat after mid-2024 and set out to find signals alive in 2024-26. Orderflow, Carry, the Seasonality rule and the combination rules came later and were chosen on dev and gate only. Counting the first version's 26 configurations as trials, the book's lockbox deflated Sharpe would be 0.16 instead of 0.20 (section 4).

The July part of the git history was rebuilt in September 2026 from the original commits. Each July commit carries the time of the last original commit it combines, and the notebooks' saved outputs are the July runs, but the contents were edited then: prose and comments were rewritten, some code was changed (notebook 01's file names and registry note among it, its July outputs kept), a few later July fixes were folded into earlier commits (the fetch's UTC dates, written on 07-07, sit in the first one), and a few pieces tied to the first version were cut; the first version itself is left out. So the freeze commit also holds a fix to Carry's funding P&L made three minutes after the freeze, before notebook 06 ran, and the research commit holds only the final wording of Seasonality's selection rule (section 3.1). The whole history was rewritten again on 2026-10-01 to change wording only, keeping every commit's date; earlier copies of this PDF were dropped from it.

Baseline sleeves (t-stat of alpha in brackets)

	Ann. return	Ann. vol	Sharpe	Max DD	Beta BTC	Beta mkt	Alpha (t)
Momentum, dev	27.9%	15.7%	+1.65	-19.6%	-0.01	-0.01	27.4% (3.5)
Momentum, gate	5.8%	15.9%	+0.43	-13.9%	-0.11	+0.01	14.3% (0.9)
Pairs, dev	-10.3%	21.7%	-0.40	-55.9%	-0.05	+0.03	-7.4% (-0.7)
Pairs, gate	8.1%	17.7%	+0.53	-10.1%	-0.04	+0.02	12.3% (0.7)

Momentum's Sharpe falls from +1.65 on dev to +0.43 on the gate. Pairs loses money on dev (-0.40), so it is dropped even though the gate was positive (+0.53). Momentum stays as the starting point for the seasonality work.

The momentum alpha needs a caveat. Alpha in this report is the intercept of a regression of daily returns on BTC and the equal-weight market, annualised. Trend following is long while prices rise and short while they fall, so its market beta keeps changing sign and averages out near zero, and almost all of its return is labelled alpha. A regression that allows for that timing (Treynor and Mazuy, on weekly returns) gives momentum a dev alpha of -0.8% a year ($t = -0.1$), with the timing term at $t = 6.8$. The Seasonality sleeve, which is built on momentum, comes out the same way (0.9%, $t = 0.3$). Their dev alphas are the payoff from riding crypto's large trends, not returns unrelated to the market.

3. Research families

Each family starts from a simple trading idea and was tested on dev and gate only. A config survives if it is positive on both windows and passes its family's extra rule: beat the untilted sleeve (seasonality), beat the unfiltered baseline (conditioned reversals), or have a neighbouring parameter that is also positive. Where more than one survives, the seasonality and order-flow notebooks keep the best gate Sharpe and the carry

notebook the best dev Sharpe, for reasons it gives. Appendix B lists all 53 registry rows.

3.1 Seasonality: who trades when

Institutions trade more on weekdays and in US hours, retail at night and at weekends. If informed flow trends, momentum should behave differently by calendar bucket. Twelve configs were tested: weekday/weekend scaling of the momentum sleeve, holding the market only in some calendar buckets, an hourly US-hours test and a turn-of-month tilt.

The survivor halves momentum's exposure on weekdays: dev +1.79 and gate +1.24, against +1.65 and +0.43 for the untilted sleeve. Almost all of the equal-weight market's return came outside US hours, but trading on that split lost money on the gate.

The neighbour rule was written two ways before any result: a neighbouring parameter that is positive on both windows, and, in a comment in the selection cell, a neighbour that is itself a survivor. No tilt meets the second, and about nine minutes after the results were logged the comment was changed to match the first (notebook 02). Without Seasonality the book's lockbox Sharpe would have been 2.11 walk-forward and 1.40 equal weight, against 1.46 and 1.23 as run.

Seasonality configs

Config	Dev Sharpe	Gate Sharpe
P1 momentum, weekend x0, weekday x1	+1.19	-0.80
P1 momentum, weekend x0.5, weekday x1	+1.47	-0.15
P1 momentum, weekend x1, weekday x1	+1.65	+0.43
P1 momentum, weekend x1.5, weekday x1	+1.75	+0.90
P1 momentum, weekend x1, weekday x0.5	+1.79	+1.24
P1 momentum, weekend x1, weekday x0	+1.34	+2.54
P2 hold market on weekdays only	+0.79	+0.64
P2 hold market at weekends only	+0.01	-1.64
P2 hold market around the turn of the month only	+0.98	-1.78
P3 hold market off hours only, hourly	+0.59	-0.35
P4 momentum, turn of month x1.5	+1.65	+0.57
P4 momentum, turn of month x2	+1.62	+0.67

The P2 and P3 rows are as run, and they held each bucket a day late (P3 an hour late): the weekday test held Tuesday to Saturday. On the days they meant they score P2 weekday +0.62 and +0.21; P2 weekend +0.29 and -1.31; P2 turn of month +0.86 and -1.74; P3 off hours +0.97 and -0.30 (dev and gate), so they still fail and the survivor stands (notebook 07, section 5).

3.2 Order flow: the taker-buy share

Binance klines report how much volume came from aggressive buyers. The uninformed-flow argument says reversal should work best on moves without real flow behind them, so reversal was tested on quiet names and on moves the flow did not confirm. The imbalance itself was also tested as a signal, in both directions.

Every reversal variant loses on dev. Following the 10-day average imbalance survives (dev +0.89, gate +2.12), and the 5-day version is also positive; the 1-day version loses on both windows. Fading the imbalance is the mirror image and loses.

Order-flow configs

Config	Dev Sharpe	Gate Sharpe
P1 reversal 3d, no filter	-0.99	+0.12
P1 reversal 5d, no filter	-1.36	+0.14

Config	Dev Sharpe	Gate Sharpe
P1 reversal 10d, no filter	-1.41	-0.69
P1 reversal 3d, volume $z < 0$	-1.05	-0.17
P1 reversal 3d, volume $z < -0.5$	-0.60	-0.27
P1 reversal 5d, volume $z < 0$	-0.88	-0.23
P1 reversal 5d, volume $z < -0.5$	-0.91	-0.77
P1 reversal 10d, volume $z < 0$	-1.31	-0.44
P1 reversal 10d, volume $z < -0.5$	-0.73	-0.77
P2 reversal 3d, flow-unconfirmed	-0.22	-0.34
P2 reversal 5d, flow-unconfirmed	-0.86	-0.11
P3 imbalance 1d, follow	-0.86	-2.77
P3 imbalance 1d, fade	-1.55	-3.21
P3 imbalance 5d, follow	+0.50	+1.45
P3 imbalance 5d, fade	-1.43	-3.60
P3 imbalance 10d, follow	+0.89	+2.12
P3 imbalance 10d, fade	-1.44	-3.60

3.3 Funding carry and crowding

Longs pay funding when it is positive, so high funding is both carry for the short side and a sign of crowded positioning. The carry sleeve shorts high-funding names and buys low-funding ones, and the backtest adds the funding payments to the price P&L.

Carry on 7-day smoothed funding survives (dev +1.01, gate +2.19), and so do its neighbours. Reversal in crowded names loses. Fading the change in funding also passes, but it is a crowding trade with a much weaker dev result, so carry was kept.

Funding configs

Config	Dev Sharpe	Gate Sharpe
P1 carry, 1d funding	+0.05	+1.75
P1 carry, 7d funding	+1.01	+2.19
P1 carry, 30d funding	+0.29	+0.72
P2 reversal 5d, funding z > 1	-0.64	-1.17
P2 reversal 5d, funding z > 1.5	-0.54	-1.43
P3 7-day change in 7d funding, follow	-1.11	-3.17
P3 7-day change in 7d funding, fade	+0.43	+2.50
P3 7-day change in 30d funding, follow	-1.02	-2.06
P3 7-day change in 30d funding, fade	+0.59	+1.59

3.4 Fast reversal (hourly)

If short-horizon reversal comes from providing liquidity, it should be stronger intraday. The rule, set before testing: a config must earn a gross return of at least 3 times its cost drag before it is worth tuning.

The 4-hour lookback has a gross Sharpe of +4.83, so the edge is real, but hourly rebalancing costs 461% a year against a gross return of 272%, about 1.7 times as much. No config comes close to the 3x rule, and net

Sharpe is negative for all eight on both windows.

Fast reversal: gross on the dev window, cost from turnover over dev and gate (hourly, 8,760 hours a year)

Lookback, rebalance	Gross Sharpe	Gross/yr	Cost/yr	Ratio	Net dev	Net gate
4h, every 1h	+4.83	272%	461%	0.59	-3.41	-7.79
4h, every 6h	+0.86	44%	146%	0.30	-1.98	-4.56
12h, every 1h	+1.80	103%	267%	0.39	-2.90	-5.05
12h, every 6h	-0.91	-50%	103%	-0.48	-2.80	-4.07
24h, every 1h	+0.63	37%	187%	0.20	-2.56	-3.37
24h, every 6h	-0.84	-48%	73%	-0.66	-2.12	-2.29
48h, every 1h	+0.01	1%	130%	0.01	-2.21	-3.44
48h, every 6h	-1.00	-57%	51%	-1.12	-1.90	-2.28

Section 5.5 reruns this grid on hourly bars for every coin that was in the top 100.

4. The book

The book holds Seasonality, Orderflow and Carry. The main version uses walk-forward max-Sharpe weights, refitted every 63 days on the previous 756 days; an equal-weight version runs alongside. Both combination rules were fixed before the lockbox and are logged in the registry.

Sleeve correlations on the dev window

	Seasonality	Orderflow	Carry
Seasonality	1.00	0.10	-0.08
Orderflow	0.10	1.00	0.07
Carry	-0.08	0.07	1.00

Book performance by window (t-stat of alpha in brackets)

	Ann. return	Ann. vol	Sharpe	Max DD	Beta BTC	Beta mkt	Alpha (t)
Walk-forward, dev	2.9%	8.1%	+0.40	-8.1%	+0.00	-0.04	3.6% (0.8)
Walk-forward, gate	21.6%	7.9%	+2.52	-3.9%	-0.05	+0.00	23.0% (2.8)
Walk-forward, lockbox	11.4%	7.6%	+1.46	-4.7%	-0.02	-0.02	9.5% (1.3)
Walk-forward, full	8.1%	7.9%	+1.02	-8.1%	-0.01	-0.02	8.3% (2.3)
Equal weight, dev	25.0%	13.8%	+1.68	-10.1%	-0.02	+0.02	22.4% (3.5)
Equal weight, gate	40.0%	11.8%	+2.90	-4.2%	-0.04	+0.02	37.2% (3.3)
Equal weight, lockbox	16.2%	12.9%	+1.23	-9.2%	-0.08	+0.03	13.4% (1.0)
Equal weight, full	25.6%	13.4%	+1.76	-10.1%	-0.02	+0.02	23.6% (4.4)

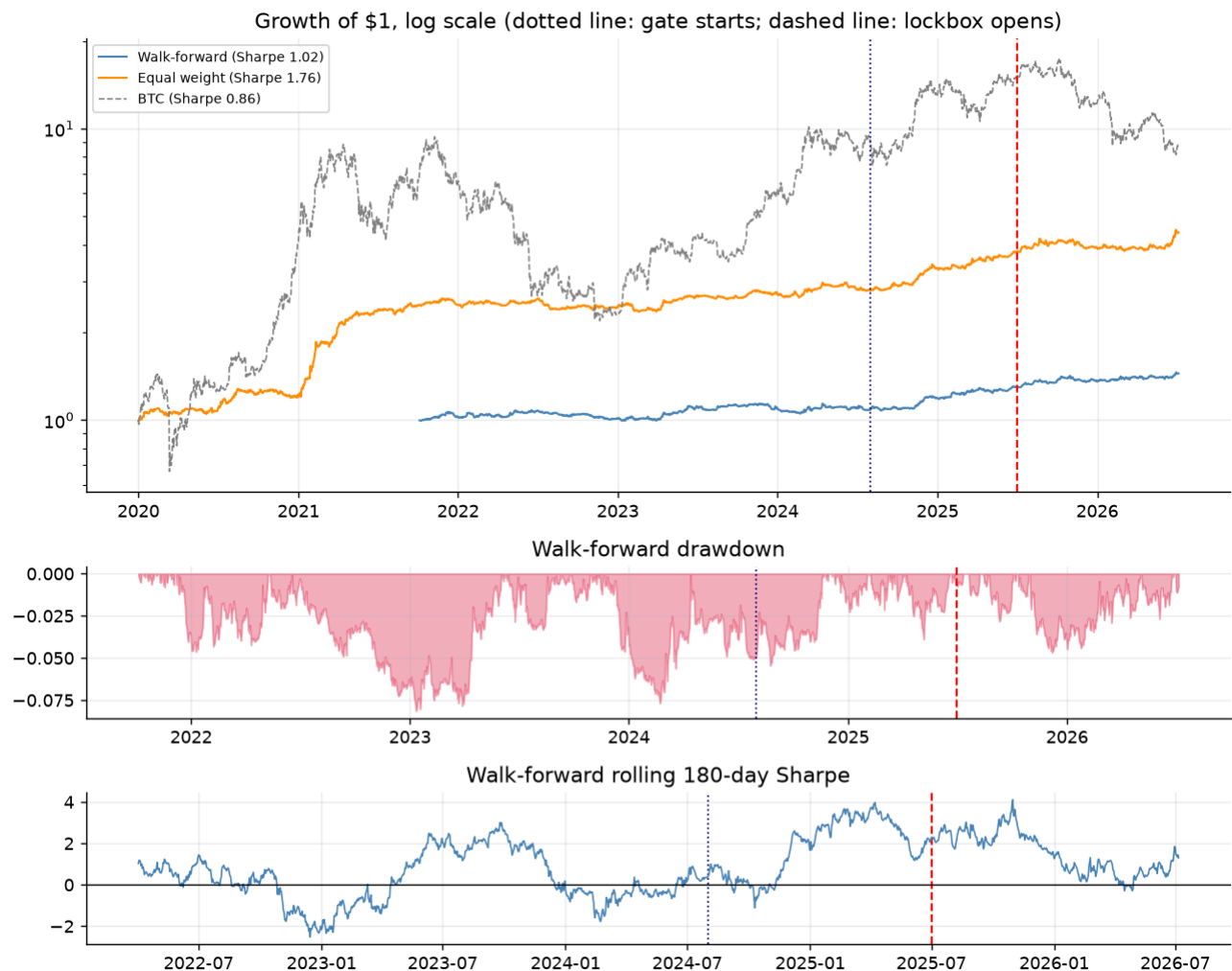
The walk-forward book needs 756 days of history before its first weights, so it starts in 2021-10 and its dev figure covers only 2021-10 to 2024-07. Equal weight beats walk-forward on every window except the lockbox: with three nearly uncorrelated sleeves there is little for mean-variance weights to add. Over the dates both books cover the margin is smaller: 0.46 against 0.40 on dev and 1.17 against 1.02 over the full sample.

Sleeve performance by window (t-stat of alpha in brackets)

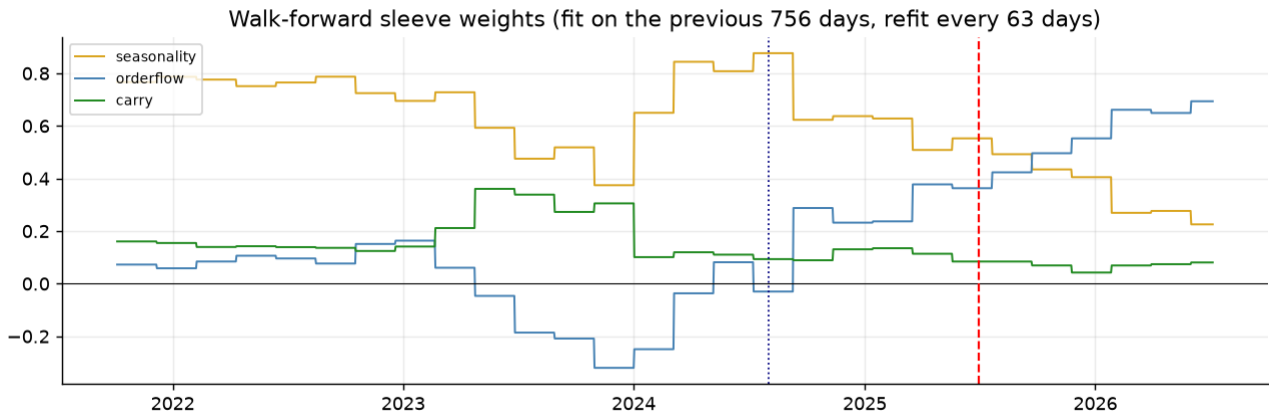
	Ann. return	Ann. vol	Sharpe	Max DD	Beta BTC	Beta mkt	Alpha (t)
Seasonality, dev	18.4%	9.7%	+1.79	-11.5%	-0.01	+0.00	18.3% (3.9)
Seasonality, gate	12.8%	10.1%	+1.24	-6.1%	-0.07	+0.00	17.0% (1.6)
Seasonality, lockbox	-5.2%	8.8%	-0.56	-8.2%	-0.05	-0.04	-9.1% (-1.2)
Orderflow, dev	20.6%	24.5%	+0.89	-29.6%	+0.01	+0.04	17.3% (1.6)

	Ann. return	Ann. vol	Sharpe	Max DD	Beta BTC	Beta mkt	Alpha (t)
Orderflow, gate	24.4%	10.6%	+2.12	-8.3%	+0.07	-0.04	18.1% (1.8)
Orderflow, lockbox	23.4%	11.1%	+1.95	-7.9%	+0.02	-0.02	22.0% (2.0)
Carry, dev	29.9%	30.5%	+1.01	-34.8%	-0.05	+0.02	31.6% (2.1)
Carry, gate	88.3%	31.1%	+2.19	-18.3%	-0.13	+0.11	76.6% (2.8)
Carry, lockbox	28.2%	35.1%	+0.88	-30.4%	-0.23	+0.14	27.2% (0.8)

The sleeve alphas look large because alpha scales with volatility. Orderflow and Carry hold \$1 of positions, half long and half short, with no volatility target, and the high-funding coins Carry trades are small and very volatile, so it runs at about 31% volatility. Scaled to 10% volatility, Carry's gate alpha of 76.6% would be 24.6%, and its lockbox alpha of 27.2% would be 7.7%. The gate is also short: Carry's three best months returned 51% together, out of 78% for the whole 11 months, and the 95% interval around its gate alpha runs from 22% to 131%. The combined walk-forward book runs at 7.6% volatility on the lockbox, with an alpha of 9.5%.



Top: growth of \$1 on a log scale. Middle: walk-forward drawdown. Bottom: rolling 180-day Sharpe of the walk-forward book.



The weights the walk-forward book used. They can go negative, and their absolute values sum to 1.

4.1 The lockbox result

Over the lockbox (371 days) the walk-forward book made 11.4% a year with 7.6% volatility and a worst drawdown of -4.7%. Its Sharpe of 1.458 has a block-bootstrap 95% interval of [-0.42, 3.44], which includes zero. Regressed on BTC and the equal-weight market it has an alpha of 9.48% a year (Newey-West $t = 1.25$) and a BTC beta of -0.020.

By sleeve, Orderflow (+1.95) and Carry (+0.88) held up, while Seasonality lost money (-0.56). Its strong gate result did not carry over, which is the risk notebook 02 flagged for an 11-month gate.

Significance

	Full sample	Lockbox
Walk-forward Sharpe	1.02	1.46
Block-bootstrap 95% CI		[-0.42, 3.44]
Deflated Sharpe (48 trials)	0.490	0.205

The deflated Sharpe is the probability that the observed Sharpe beats the best Sharpe that 48 strategies with no real edge would show by luck. It is 0.49 on the full sample and 0.20 on the lockbox (0.16 counting the first version's 26 configurations, section 2), both far below 0.95. One year is also short: the bootstrap interval above is 3.9 Sharpe points wide. The book did make money with almost no market exposure, but this sample cannot separate that from luck. Section 5 reruns the same book on a universe with every pair, where its lockbox Sharpe is +0.35.

4.2 The whole sample, as run

Results over the whole sample (2020-01-01 to 2026-07-06), research coin list

	From	Ann. return	Ann. vol	Sharpe	Max DD	Beta BTC	Beta mkt	Alpha (t)
Baseline momentum	2020-01	18.4%	15.5%	+1.17	-19.6%	-0.02	-0.02	20.2% (3.2)
Baseline pairs	2020-01	-8.0%	20.2%	-0.31	-55.9%	-0.04	+0.02	-5.4% (-0.7)
Seasonality	2020-01	13.6%	9.6%	+1.37	-11.5%	-0.01	-0.01	14.5% (3.8)
Orderflow	2020-01	21.5%	21.3%	+1.02	-29.6%	+0.02	+0.02	19.6% (2.4)
Carry	2020-01	36.6%	31.3%	+1.15	-35.6%	-0.08	+0.05	36.6% (2.9)
Book, walk-forward	2021-10	8.1%	7.9%	+1.02	-8.1%	-0.01	-0.02	8.3% (2.3)
Book, equal weight	2020-01	25.6%	13.4%	+1.76	-10.1%	-0.02	+0.02	23.6% (4.4)
BTC	2020-01	39.3%	60.8%	+0.86	-76.6%			
Equal-weight market	2020-01	32.3%	83.8%	+0.77	-84.8%			

Section 5 shows how little of this survives a universe with every pair. Alpha and the two betas come from a regression of daily returns on

BTC and the equal-weight market; the *t*-stat in brackets uses Newey-West errors. These figures mix the windows the strategies were chosen on with the windows they were tested on. Section 2 explains why the momentum alphas are mostly market timing, and the start of this section why Carry's alpha is large.

5. Making the backtest realistic

Notebooks 09 and 11 were added on 2026-09-27, after the forward test (section 6). Notebook 09 rebuilds the daily data from Binance's public archive, which keeps the files of delisted pairs, and checks it against the research data: every price and funding rate the two have in common is the same. Notebook 11 reruns the frozen book with one assumption at a time made more realistic. Like appendix A, it re-scores the lockbox with corrected inputs and chooses nothing on it, and v2 (section 7) was registered before notebook 11 was first run.

5.1 The coin list

The research coin list is the 150 USDT pairs with the most 24-hour volume on the day of the fetch, 2026-07-06, so it holds mostly coins that were still big then, plus 11 that had in fact stopped trading. From 2020 to July 2026, 46% of the universe's coin-days belonged to coins it left out: coins that were delisted or renamed, such as EOS, MATIC, XMR, WAVES and MKR, and coins that had just shrunk, such as VET, SAND, GRT, KAVA and EGLD. The share is highest in the early years, when the list was furthest from its fetch date:

Share of the daily top-100 universe outside the research coin list

Year	2020	2021	2022	2023	2024	2025	2026
Coin-days the list did not have	55%	60%	53%	51%	41%	33%	23%

The list had three smaller faults. Its filter for leveraged tokens dropped any name ending in UP, which removed JUP and SYRUP. Six pegged assets got in: the euro, the gold tokens PAXG and XAUT, and three newer dollar tokens. EUR and PAXG sat in the universe for most of the sample, and Carry could trade the two gold perps. And five tokenized US stocks, which Binance began listing in June 2026, entered the universe during the forward test. The archive needed care too: it keeps writing a frozen, zero-volume bar every day for a perp that has been delisted, and some perps stop tracking their coin after a token migration. So a strategy only picks a perp on days the contract traded and priced within 20% of spot, while a position already held earns its contract's own move and funding on every day the contract traded (section 5.2).

5.2 The frozen book on corrected data

The frozen book, one fix at a time (Sharpe; each step keeps the ones before it)

Step	WF dev	WF gate	WF lockbox	WF fwd	EW dev	EW gate	EW lockbox	EW fwd
0 As run	+0.40	+2.52	+1.46	-1.01	+1.68	+2.90	+1.23	+0.63
0b Same list, archive	+0.40	+2.52	+1.45	-1.01	+1.68	+2.90	+1.23	+0.63
1 Non-crypto out	+0.68	+2.21	+1.26	-1.11	+1.55	+2.83	+1.20	+0.57
2 Every pair	-0.20	+1.85	+0.35	-0.90	+0.79	-1.44	-0.62	+1.26
3 Carry on perps	-0.33	+1.71	-0.01	-0.88	+0.80	-1.20	-0.46	+1.46
4 Limit fills	-0.41	+2.15	+0.12	-0.31	+0.58	-1.43	-0.69	+1.43

Rebuilt from the archive, the research list gives the same numbers (step 0b), so the data source is not the issue. Taking out the non-crypto assets costs a little: the walk-forward book's lockbox Sharpe goes from 1.46 to 1.26. Step 2 is the one that matters. With every pair in the universe the walk-forward book's lockbox Sharpe is +0.35 and the equal-weight book's -0.62, and the equal-weight book loses money on the gate as well (-1.44). The walk-forward numbers also depend on the book's refit dates: started up to eight weeks later, the research book's lockbox Sharpe ranges from 1.30 to 1.55 and the every-pair book's from 0.08 to 0.57. The gap between them holds for any start; the equal-weight book has no refit dates. The every-pair books start on the research's first Carry day, 2019-09-10, so that they refit on the research's dates; every-pair funding only

starts on 2020-01-01, so Carry is flat until then. The first run of notebook 11 started them in 2018, with nearly two years of flat Carry, and put the every-pair walk-forward lockbox Sharpe at 0.16.

The three sleeves at steps 0, 2 and 4 (Sharpe)

Sleeve, step	Dev	Gate	Lockbox	Forward
Seasonality, as run	+1.79	+1.24	-0.56	+1.17
Seasonality, every pair	+1.57	+1.42	-0.65	+1.96
Seasonality, limit fills	+1.57	+1.42	-0.65	+1.96
Orderflow, as run	+0.89	+2.12	+1.95	-2.56
Orderflow, every pair	+0.44	-0.78	-0.40	-1.44
Orderflow, limit fills	+0.09	-1.34	-0.92	-1.61
Carry, as run	+1.01	+2.19	+0.88	+0.90
Carry, every pair	+0.25	-1.82	-0.42	+1.57
Carry, limit fills	+0.19	-1.56	-0.35	+1.76

Orderflow and Carry lose money on the gate and the lockbox once every pair is in. Seasonality stays positive on dev and gate (+1.57 and +1.42, against +1.79 and +1.24 as run) and still loses on the lockbox. The coins that did the damage are mostly ones the research list never had. On the gate, four of Orderflow's five worst coins were missing from it, OM above all, which collapsed in April 2025, and so were all five of Carry's: delisting casualties such as VIDT and BNX, which the z-score weights bought because their funding had turned negative. Measuring Carry on perp prices (step 3) helps it a little on the gate and the lockbox. Two things were fixed on the way. The archive's monthly perp files for February and April 2022 are missing days for about 50 contracts (the last three of February, the first two of April), and notebook 09 now fills them from the archive's daily files. And on a day the 20% rule drops, the first run gave a coin Carry already held no funding (steps 2 and 3) and no price move (step 3). A held position earns what its contract did. On a few of those days the perp still traded, however far from spot, as in LUNA's and FTT's crashes in 2022 and OMG's steep discount to spot in November 2021, and its own move and funding now count; on the rest it did not trade, mostly because it had been delisted, and the position earns nothing. With those days at zero, Carry's step-3 Sharpe would read +0.37 on dev and -1.58 on the gate instead of +0.25 and -1.51.

5.3 Limit orders that have to fill

The research charges 7 bps and assumes every limit order fills at the close. In step 4 an order placed at the close only fills if the next day trades through its price, and a missed order is replaced at the next close. Almost all of them fill:

Limit orders on dev, on coins that traded the next day (P&L in points: sums of daily price returns, Carry's before funding)

	Orders filled	Positions missed	Return, missed buy	Return, missed sell	P&L, all filled	P&L, fill model
Orderflow	99.0%	0.29%	7.1%	-5.7%	69.9	39.8
Carry	99.5%	0.11%	7.5%	-5.5%	-136.6	-147.3

The orders that miss are the ones the price ran away from. Where a buy missed, the coin rose 7.1% that day on average, and where a sell missed it fell 5.7%. Missing 0.3% of its positions costs Orderflow 30 of its 70 points of price P&L on dev, and its dev Sharpe falls from 0.44 to 0.09. Carry trades less and loses less. Its orders fill on its perps' own bars, whatever their basis to spot. An order cannot fill on a day the contract does not trade; if it trades again within three days, a position left open takes the move across the gap, and after more than three days the contract counts as delisted: the position is settled at its last price and dropped, so a relaunch under the same name, like LUNA's in 2022, cannot revive it. A day's funding is credited to the position after that day's fill, although the prints before a fill belong to the old position; daily funding data cannot split the day.

5.4 Size and capacity

The same Orderflow rule on the largest coins and on the smaller ones shows where the research list's edge came from:

Orderflow by coin size (Sharpe)

Universe, coins	Dev	Gate	Lockbox	Forward
Research list, top 30	+0.92	-0.19	-0.36	-0.51
Research list, top 50	+0.78	+1.15	+0.45	-1.58
Research list, ranks 51-100	+0.11	+1.58	+1.66	-2.17
Research list, top 100	+0.70	+2.36	+1.75	-2.43
Every pair, top 30	+0.97	+0.71	-0.18	+0.06
Every pair, top 50	+1.05	+0.51	+0.33	-1.77
Every pair, ranks 51-100	+0.00	-2.14	-1.01	+0.51
Every pair, top 100	+0.44	-0.78	-0.40	-1.44

On the research list, Orderflow's gate and lockbox results come from the coins ranked 51 to 100 (+1.58 and +1.66). On every pair those coins lose money on both (-2.14 and -1.01). The research list's small coins were the ones that went on to be big in 2026, which is how survivorship shows up in a signal.

Market impact is modelled per coin with the square-root law: trading Q dollars of a coin with daily volume V and daily volatility σ costs about $y \times \sigma \times \sqrt{Q/V}$ on top of the 7 bps fee, with y between 0.5 and 1. Binance volume alone makes both settings on the harsh side. On the research-list sleeves, over 2020 to July 2026:

Research-list sleeves with square-root market impact (Sharpe)

Money run	Orderflow, $y = 0.5$	Orderflow, $y = 1$	Carry, $y = 0.5$	Carry, $y = 1$
no impact	+0.87	+0.87	+1.13	+1.13
\$0.1M	+0.69	+0.51	+1.05	+0.96
\$1M	+0.30	-0.27	+0.86	+0.59
\$10M	-0.92	-2.55	+0.27	-0.57
\$50M	-2.90	-5.53	-0.76	-2.31

Even taken at face value, the research-list Orderflow could not have run much money: its Sharpe falls from 0.87 to 0.30 at \$1M with $y = 0.5$, and below zero with $y = 1$. Carry holds up to somewhere between \$1M and \$10M.

5.5 Hourly reversal on every pair

Notebook 05's hourly test (section 3.4) ran on 60 coins picked by volume in July 2026, about 29 of them tradable in a typical hour. Notebook 11 reruns its eight configs, with the same rules and costs, on hourly bars for every coin that was in the daily top 100, about 96 an hour. On the 60 coins it reproduces notebook 05 exactly.

Notebook 05's grid on its 60 coins and on every pair (dev, 8,760 hours a year)

Lookback, rebalance	Gross Sharpe, 60 coins	Gross Sharpe, every pair	Gross/cost, 60 coins	Gross/cost, every pair	Net dev, every pair	Net gate, every pair
4h, every 1h	+4.83	+6.92	0.59	0.99	-0.09	-5.77
4h, every 6h	+0.86	+1.70	0.30	0.68	-0.81	-2.79
12h, every 1h	+1.80	+4.31	0.39	0.97	-0.18	-3.88
12h, every 6h	-0.91	-0.03	-0.48	-0.02	-1.84	-2.52

Lookback, rebalance	Gross Sharpe, 60 coins	Gross Sharpe, every pair	Gross/cost, 60 coins	Gross/cost, every pair	Net dev, every pair	Net gate, every pair
24h, every 1h	+0.63	+2.40	0.20	0.75	-0.81	-2.54
24h, every 6h	-0.84	-0.16	-0.66	-0.13	-1.45	-1.64
48h, every 1h	+0.01	+1.55	0.01	0.64	-0.87	-2.32
48h, every 6h	-1.00	-0.47	-1.12	-0.48	-1.45	-1.42

With the wider universe the gross edge is larger, which is the opposite of what the daily sleeves showed: the 4-hour lookback rebalanced every hour has a gross Sharpe of 6.92 against 4.83, and its gross return is about equal to its trading cost (0.99 times). That is still far from the rule of 3 times the cost, and every config loses money after costs on dev and on the gate, so the conclusion of section 3.4 stands.

5.6 The pairs baseline, fully charged and without its look-ahead

Charging the pairs engine for every position carried across a rebalance, an upper bound on what it missed, moves its dev Sharpe from -0.40 to -0.43, its gate Sharpe from +0.53 to +0.48 and its lockbox Sharpe from -0.69 to -0.75. The engine also picked its pairs with the close of the rebalance date and then booked that date's return for them, a one-day look-ahead at each rebalance. Picking them with data up to the day before gives -0.37, +0.43 and -0.72 on dev, gate and lockbox, and with both changes -0.41, +0.39 and -0.78. Pairs loses money on dev either way.

6. Out-of-sample evidence

The frozen book has two stretches of data it never influenced: the lockbox year and the 82 days after the research data was fetched.

6.1 The forward test, July to September 2026

Notebook 08, added on 2026-09-27, downloads the days after the research data ends for the same coins and runs the frozen book on them, with the same universe rule, sleeves, costs and walk-forward settings. Where the new download overlaps the cached data (May to early July 2026) every daily bar and funding rate matches, and up to 2026-07-05 the rebuilt sleeves and books match the research exactly. Nothing was tuned and no trials were logged. The window runs from 2026-07-07 to 2026-09-26, 82 days.

The frozen book on the forward window (82 days; return is the total over the window, beta is to BTC alone)

	Return	Ann. vol	Sharpe	Max DD	Beta BTC alone
Book, walk-forward	-5.9%	24.3%	-1.01	-9.4%	-0.12
Book, equal weight	5.1%	59.4%	+0.63	-11.3%	-0.20
Walk-forward, resizing charged	-7.3%	27.9%	-1.08	-11.2%	-0.15
Equal weight, resizing charged	4.7%	59.4%	+0.61	-11.4%	-0.20
Seasonality	2.6%	10.0%	+1.17	-3.5%	+0.07
Orderflow	-10.8%	19.1%	-2.56	-13.5%	-0.12
Carry	11.3%	166.6%	+0.90	-26.3%	-0.55
BTC	31.8%	40.3%	+3.25	-6.9%	
Equal-weight market	37.2%	51.1%	+3.01	-11.1%	

The walk-forward book lost 5.9% while BTC gained 31.8%. Orderflow, with 69% to 80% of the book's weight, lost 10.8% after being the best sleeve on the lockbox. It has had runs like this before: 4.9% of its 82-day stretches since 2020 were worse, about one in twenty. Some of the loss was the market. Over this window the book's beta to BTC alone was -0.12, against -0.04 over the research period, and beta times BTC's rise accounts for 3.5 of the 5.5 points the book lost, counting in sums of daily returns.

The equal-weight book made 5.1% only because of Carry, and Carry's 11.3% came from one coin. DEXE fell

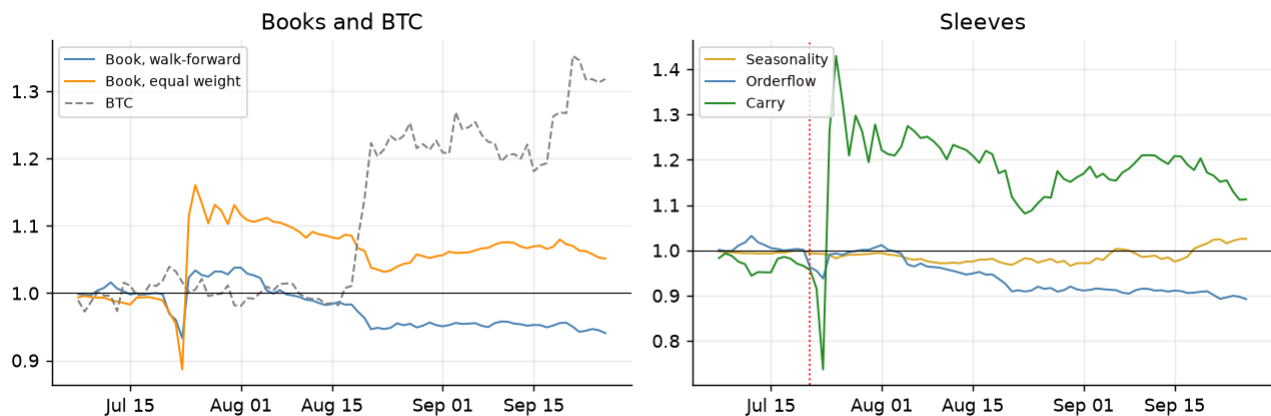
82.5% on 2026-07-21 and swung hard for the next week. Traders shorted its perp heavily and funding went deeply negative: shorts paid longs 13% to 21% a day for three days. Carry buys low-funding coins, and its z-score weights have no cap per coin, so from 2026-07-22 it held DEXE at 47% to 50% of its gross for 12 days, all or nearly all of its long side. Counted in sums of daily returns, DEXE added 37.7 points; all the other coins together lost 2.8, and trading costs took 1.1. The gain does not come from pairing perp funding with spot prices. The perp closed up to 10.5% below spot on the worst days and the gap closed while Carry held it: from 2026-07-18 to 2026-08-03, the price leg on the perp itself was 12.6 points against 0.8 on spot, next to 40.9 from funding. What makes the gain fragile is that one coin supplied it.

Carry: the largest single-coin weight each day, as a share of gross exposure

Window	Days	Median largest weight	Share of days ≥ 0.25	Days one coin held a whole side
Dev	1674	0.20	32%	23
Gate	334	0.19	27%	0
Lockbox to 2026-07-05	370	0.17	27%	0
Forward	82	0.27	59%	10

The weighting always allowed this. In the research, one coin held a whole side on 23 days, all on dev: 16 in January 2020, when only 3 to 9 coins had funding data, and 7 in November 2022, when the sleeve held SOL through the FTX collapse. The forward window had 10 such days, all DEXE.

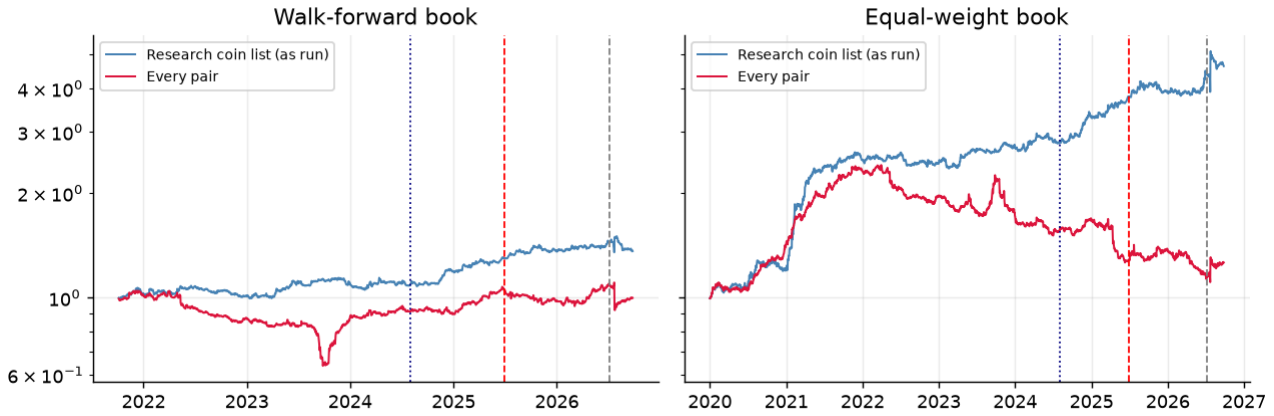
An 82-day Sharpe has a standard error of about 2.1, so this window can neither confirm nor overturn the lockbox. It does show, more plainly than the research did, that Carry's uncapped weights can put half the sleeve into one coin in the middle of a crash. A per-coin cap is the obvious fix, but a cap chosen after seeing this window would be fitted to it, so the book is reported as it was frozen. Section 6.2 adds this window to the lockbox.



Growth of \$1 over the forward window. Left: the two books and BTC. Right: the three sleeves; the dotted line marks DEXE's crash on 2026-07-21.

6.2 The whole out-of-sample record

The lockbox year and the forward test are the only data the frozen book never influenced. Together, as run, they give the walk-forward book a Sharpe of 0.37 over 453 days ($t = 0.4$; block-bootstrap 95% interval -1.22 to 1.95) and the equal-weight book 0.72. The rest of 2026-07-06 after the research data was fetched falls in neither window (-0.03% for the walk-forward book). In the forward window the walk-forward book's beta to BTC was lower than in 93% of the earlier 82-day stretches, and Orderflow's lower than in 99%.



Growth of \$1 on a log scale for the frozen book as run (research coin list) and on every pair. Dotted line: gate starts; dashed red: lockbox opens; dashed grey: forward test starts.

7. Version 2, registered before testing

v2 is the book rebuilt around what the audits found. Its rules were committed to the repository on 2026-09-27, in notebook 10 and strategies.py, before any v2 number existed, and the next commit adds the results. It makes five changes, each answering a problem the audits found or a result of the frozen book on dev and gate, none chosen on v2's own results. One of those problems showed up in the forward window: the DEXE case of section 6.1, which rank weights also answer. It drops Seasonality, which fails notebook 02's rule once its resizing trades are charged. It weights Carry by rank instead of z-score, so once more than a handful of coins have funding data no coin can take a whole side of the sleeve. It measures Carry on perp prices. It trades the every-pair universe. And it combines its two sleeves with equal weights, which beat the walk-forward weights on dev (0.46 against 0.40 over the same dates) and on the gate (2.90 against 2.52). v2 is never evaluated on the lockbox or on the forward-test window, since both have been seen; notebook 12 uses those months only as history for v2's first positions.

Two fixes came after v2's first run, the same two as in section 5.2. A Carry position held into a day the 20% rule drops its perp's data now earns what its contract did, its own move and funding, or nothing once the contract has stopped trading (strategies.v2_sleeves with pnl_funding and the archive's traded panels). And the two 2022 holes in the archive are filled. Neither changes v2's rules. The first changes what v2 earns on a few days; the second gives Carry funding data it was missing on those days of 2022, which also changes its positions for about a week after each hole. Both were made on 2026-09-29 (UTC), before any test result was computed and before the test began. An earlier version of the first, committed on 2026-09-28, gave a held coin its spot move instead, which is wrong both on days the perp trades far from spot and after a delisting. The table uses both fixes. As first run, v2 had a Sharpe of 1.39 on dev and 1.12 on the gate, and its Carry 1.76 and 2.41; the registry and notebook 10 keep those numbers.

v2 on dev and gate, every pair (t-stat of alpha in brackets)

	Ann. return	Ann. vol	Sharpe	Max DD	Beta BTC	Beta mkt	Alpha (t)
v2 book, dev	17.5%	12.6%	+1.35	-23.0%	+0.03	-0.04	17.7% (2.9)
v2 book, gate	11.6%	10.2%	+1.13	-11.2%	+0.04	-0.05	6.9% (0.7)
Orderflow, dev	6.7%	18.3%	+0.44	-29.7%	+0.05	-0.06	8.9% (1.1)
Orderflow, gate	-11.2%	14.0%	-0.78	-22.9%	+0.05	-0.04	-15.8% (-1.0)
Carry, dev	27.8%	15.3%	+1.68	-24.6%	+0.02	-0.03	26.5% (3.2)
Carry, gate	39.1%	14.0%	+2.42	-7.8%	+0.04	-0.05	29.7% (2.0)

On dev and gate v2 has a Sharpe of 1.35 and 1.13, and nearly all of it comes from Carry: weighted by rank, on perp prices and across every pair, Carry has 1.68 and 2.42, at about half the frozen Carry's volatility (14% against 31% on the gate). Orderflow does not survive the wider universe (+0.44 on dev, -0.78 on the gate), so

by notebook 03's rule it would not have been kept; it stays because v2's rules were fixed before this check. The three rows are in the registry.

The test is every day from 2026-10-02. The rules and both fixes were first pushed to GitHub on 2026-09-29 (15:17 UTC); the history was rewritten for wording on 2026-10-01 and force-pushed, which replaced those commits, so the test starts on the first full day after that push. The start was first set to 2026-09-28, the day after the rules were committed, and has only moved later, before any test day was computed, because a commit's date is only evidence for others once it is public. Notebook 12 adds each month from the archive, delisted coins included, and reports v2's book and each sleeve alongside the frozen book. At a Sharpe near 1.5, a t-stat of 2 takes about two years of data. Until then v2 is a hypothesis under test, not a validated strategy: its dev and gate figures come from windows that had been seen when it was designed.

8. The sleeves in twsq

alphas/ holds the three sleeves written as alphas for twsq, an execution framework. twsq backtests them on Binance daily bars over 701 days (2024-08-06 to 2026-07-07) with the same costs. They differ from the research in ways that matter: they trade a fixed list of 20 large coins instead of the point-in-time top 100, twsq charges every trade (including the Seasonality resizing), and the carry alpha holds spot, so it earns no funding. The signals only use data up to the previous day's close.

twsq backtests, 2024-08-06 to 2026-07-07 (daily P&L on a fixed \$1M; the annual return is the daily mean x 365)

Alpha	Ann. return	Ann. vol	Sharpe	Max DD	Turnover/day	Fees, total
SeasonalMomentum	0.7%	9.9%	+0.07	-10.9%	8%	3.7%
OrderflowFollow	-2.3%	15.5%	-0.15	-27.6%	29%	14.0%
FundingCarry	3.8%	17.2%	+0.22	-20.6%	21%	10.3%

On 20 large coins none of the three does much. SeasonalMomentum is roughly flat, which fits the charged Seasonality numbers in appendix A.1. OrderflowFollow loses a little; section 5.4 shows why, since the research list's order-flow edge came from its smaller coins, and those were survivors. FundingCarry's price leg is slightly positive, without the funding income that the research sleeve earns.

9. Limitations

- One exchange. Prices, volumes, taker flow and funding all come from Binance, and taker flow has no second source here.
- Short samples. The lockbox is one year and the forward test 82 days. The walk-forward book's deflated Sharpe is below 0.95 on every window, and on the lockbox both books are far below it (0.20 walk-forward, 0.16 equal weight). v2 has no out-of-sample data yet.
- Approximate models. The fill model works from daily highs and lows, and the market-impact model is a rough guide that uses Binance volume only.
- Free shorts. The spot sleeves (momentum, Orderflow, pairs) short coins without paying to borrow them; backtest.run can charge a borrow rate, but Binance's margin rates are not in the data.
- A hand-kept list. Tokenized stocks are excluded by a list in data.TOKENIZED_STOCKS; notebook 12 flags any new listing that looks like one.

Appendix A: Checks from the first audit

Notebook 07 was added in September 2026, after the lockbox had been opened. It changes nothing in the book and logs no trials; it measures the things the audit raised and adds the whole-sample figures in section 4.2.

A.1 The cost of resizing the Seasonality sleeve

The Seasonality overlay multiplies each day's return by 0.5 on weekdays and 1.0 at weekends. That is the same as trading down to half size every Monday and back up every Saturday, but the backtest never charged

for those trades. Rebuilding the positions actually held and charging 20 bps on every change costs the sleeve 2.7% a year on dev:

Seasonality with and without the resizing cost (Sharpe)

	Dev	Gate	Lockbox
Seasonality, as run	+1.79	+1.24	-0.56
Seasonality, every trade charged	+1.51	+0.99	-0.89
Untilted momentum, every trade charged	+1.64	+0.42	not run

Charged properly, the overlay still beats untitled momentum on the gate but not on dev, so it would have failed notebook 02's rule. The error is in the cost accounting; the signal itself is computed correctly. The frozen book was left alone, and the whole book was re-scored with the charged sleeve instead:

The frozen book, as run and with the resizing charged

	As run	Resizing charged
Walk-forward Sharpe, dev	0.40	-0.10
Walk-forward Sharpe, gate	2.52	2.51
Walk-forward Sharpe, lockbox	1.46	1.49
Walk-forward Sharpe, full	1.02	0.70
Equal weight Sharpe, dev	1.68	1.62
Equal weight Sharpe, gate	2.90	2.83
Equal weight Sharpe, lockbox	1.23	1.16
Equal weight Sharpe, full	1.76	1.70
Walk-forward lockbox 95% CI	[-0.42, 3.44]	[-0.40, 3.46]
Deflated Sharpe, full	0.490	0.234
Deflated Sharpe, lockbox	0.205	0.214
Walk-forward lockbox beta to BTC	-0.020	-0.016

On the lockbox the change is small and goes both ways. The walk-forward book gains slightly: with the weaker charged sleeve in its training data it gives Seasonality less weight (an average of 0.30 over the lockbox, against 0.37 as run), and Seasonality lost money on the lockbox. The equal-weight book loses a little. Over the full sample the walk-forward Sharpe drops from 1.02 to 0.70. The same shortcut applies wherever returns are scaled by a volatility target, but there it is small: charging it moves plain momentum's dev Sharpe by 0.01.

A.2 Higher costs for Orderflow and Carry

Sharpe at higher costs (dev and gate only)

Cost per dollar traded	Orderflow dev	Orderflow gate	Carry dev	Carry gate
7 bps	+0.89	+2.12	+1.01	+2.19
14 bps	+0.61	+1.38	+0.84	+2.03
20 bps	+0.37	+0.75	+0.70	+1.90

Both sleeves assume their daily rebalancing fills as limit orders at 7 bps. If every trade had to cross the spread at 20 bps, both would stay positive, but Orderflow would lose 58% of its dev Sharpe; it trades 26% of its book a day on dev. Carry trades 20% a day and earns much of its P&L from funding, so it is less exposed to costs. Holding constant targets also takes small daily trades as positions drift with prices, which no backtest here charges; at 7 bps they would cost Orderflow 0.2% a year on dev and Carry 0.2% (notebook 07, section 6). Nor do the books' daily moves of money between their sleeves get charged: before any netting between sleeves, they would cost the equal-weight book 0.14% a year on dev and the walk-forward book 0.07%.

A.3 Where the carry P&L comes from

Carry sleeve: price and funding legs (arithmetic annual means)

	Price leg/yr	Funding leg/yr	Funding share	Price Sharpe	Funding Sharpe
Dev	11.4%	19.3%	63%	0.38	4.79
Gate	41.2%	26.9%	39%	1.32	12.34

On dev most of the carry return is funding income, and the funding leg on its own is very smooth. On the gate the price leg did more of the work. A spot-only version of the trade, like the twsq one in section 8, earns only the price leg.

Appendix B: Trial registry

Every configuration tested, in the order it was run, with its dev (2020-01-01 to 2024-07-31) and gate (2024-08-01 to 2025-06-30) Sharpe. Rows 1 to 48 are the research trials used for the deflated Sharpe; rows 49 and 50 are the two combination rules for the book, and rows 51 to 53 are v2, logged after it was registered (section 7).

#	Family	Config	Dev	Gate
1	baseline	momentum, 30d lookback, 15% vol target	+1.65	+0.43
2	baseline	pairs, entry 2 sigma, 30d z-score, top 20	-0.40	+0.53
3	seasonality	P1 momentum, weekend x0, weekday x1	+1.19	-0.80
4	seasonality	P1 momentum, weekend x0.5, weekday x1	+1.47	-0.15
5	seasonality	P1 momentum, weekend x1, weekday x1	+1.65	+0.43
6	seasonality	P1 momentum, weekend x1.5, weekday x1	+1.75	+0.90
7	seasonality	P1 momentum, weekend x1, weekday x0.5	+1.79	+1.24
8	seasonality	P1 momentum, weekend x1, weekday x0	+1.34	+2.54
9	seasonality	P2 hold market on weekdays only	+0.79	+0.64
10	seasonality	P2 hold market at weekends only	+0.01	-1.64
11	seasonality	P2 hold market around the turn of the month only	+0.98	-1.78
12	seasonality	P3 hold market off hours only, hourly	+0.59	-0.35
13	seasonality	P4 momentum, turn of month x1.5	+1.65	+0.57
14	seasonality	P4 momentum, turn of month x2	+1.62	+0.67
15	orderflow	P1 reversal 3d, no filter	-0.99	+0.12
16	orderflow	P1 reversal 5d, no filter	-1.36	+0.14
17	orderflow	P1 reversal 10d, no filter	-1.41	-0.69
18	orderflow	P1 reversal 3d, volume z < 0	-1.05	-0.17
19	orderflow	P1 reversal 3d, volume z < -0.5	-0.60	-0.27
20	orderflow	P1 reversal 5d, volume z < 0	-0.88	-0.23
21	orderflow	P1 reversal 5d, volume z < -0.5	-0.91	-0.77
22	orderflow	P1 reversal 10d, volume z < 0	-1.31	-0.44
23	orderflow	P1 reversal 10d, volume z < -0.5	-0.73	-0.77
24	orderflow	P2 reversal 3d, flow-unconfirmed	-0.22	-0.34
25	orderflow	P2 reversal 5d, flow-unconfirmed	-0.86	-0.11
26	orderflow	P3 imbalance 1d, follow	-0.86	-2.77

#	Family	Config	Dev	Gate
27	orderflow	P3 imbalance 1d, fade	-1.55	-3.21
28	orderflow	P3 imbalance 5d, follow	+0.50	+1.45
29	orderflow	P3 imbalance 5d, fade	-1.43	-3.60
30	orderflow	P3 imbalance 10d, follow	+0.89	+2.12
31	orderflow	P3 imbalance 10d, fade	-1.44	-3.60
32	carry	P1 carry, 1d funding	+0.05	+1.75
33	carry	P1 carry, 7d funding	+1.01	+2.19
34	carry	P1 carry, 30d funding	+0.29	+0.72
35	carry	P2 reversal 5d, funding z > 1	-0.64	-1.17
36	carry	P2 reversal 5d, funding z > 1.5	-0.54	-1.43
37	carry	P3 7-day change in 7d funding, follow	-1.11	-3.17
38	carry	P3 7-day change in 7d funding, fade	+0.43	+2.50
39	carry	P3 7-day change in 30d funding, follow	-1.02	-2.06
40	carry	P3 7-day change in 30d funding, fade	+0.59	+1.59
41	fast reversal	4h lookback, rebalance every 1h	-3.41	-7.79
42	fast reversal	4h lookback, rebalance every 6h	-1.98	-4.56
43	fast reversal	12h lookback, rebalance every 1h	-2.90	-5.05
44	fast reversal	12h lookback, rebalance every 6h	-2.80	-4.07
45	fast reversal	24h lookback, rebalance every 1h	-2.56	-3.37
46	fast reversal	24h lookback, rebalance every 6h	-2.12	-2.29
47	fast reversal	48h lookback, rebalance every 1h	-2.21	-3.44
48	fast reversal	48h lookback, rebalance every 6h	-1.90	-2.28
49	book	walk-forward mean-variance, 756d fit, 63d step	+0.40	+2.52
50	book	equal weight	+1.68	+2.90
51	v2	Orderflow, every pair	+0.44	-0.78
52	v2	Carry, rank weights on perps, every pair	+1.76	+2.41
53	v2	v2 book, equal weight	+1.39	+1.12